Assignment 02

1. Significant earthquakes since 2150 B.C.

1.1

Before performing the statistical analysis, missing values in the dataset were replaced with zeros. The top 20 countries ranked by total number of deaths are presented below.

Table 1 The Top 20 Countries

	Country	Total Deaths
1	CHINA	2106526
2	TURKEY	1008867
3	IRAN	761656
4	SYRIA	488326
5	ITALY	417297
6	HAITI	323782
7	JAPAN	319510
8	AZERBAIJAN	319251
9	INDONESIA	283485
10	ARMENIA	175000
11	PAKISTAN	143714
12	ECUADOR	134445
13	TURKMENISTAN	110412
14	PERU	97009
15	GREECE	83658
16	PORTUGAL	82572
17	IRAQ	70200
18	CHILE	70175
19	INDIA	63413
20	ISRAEL	58368

1.2

As shown in Figure 1, the number of recorded earthquakes remained low (0–3) before 1500. From 1500 to 2000, despite some fluctuations, the overall trend showed a gradual increase. After 2000, the number of recorded earthquakes declined sharply.

The observed pattern may be attributed to changes in historical record-keeping, seismic monitoring technology, and data completeness. Before 1500, limited written records likely resulted in substantial underreporting of earthquakes. From 1500 to 2000, improvements in historical documentation and the development of seismic monitoring networks may have contributed to the increase in recorded earthquake counts. In contrast, the sharp decline after 2000 may reflect incomplete data for recent years rather than a genuine decrease in earthquake occurrence. Therefore, this trend should be interpreted cautiously, as variations in recorded earthquake counts may reflect changes in data availability and recording practices rather than actual changes in seismic activity.

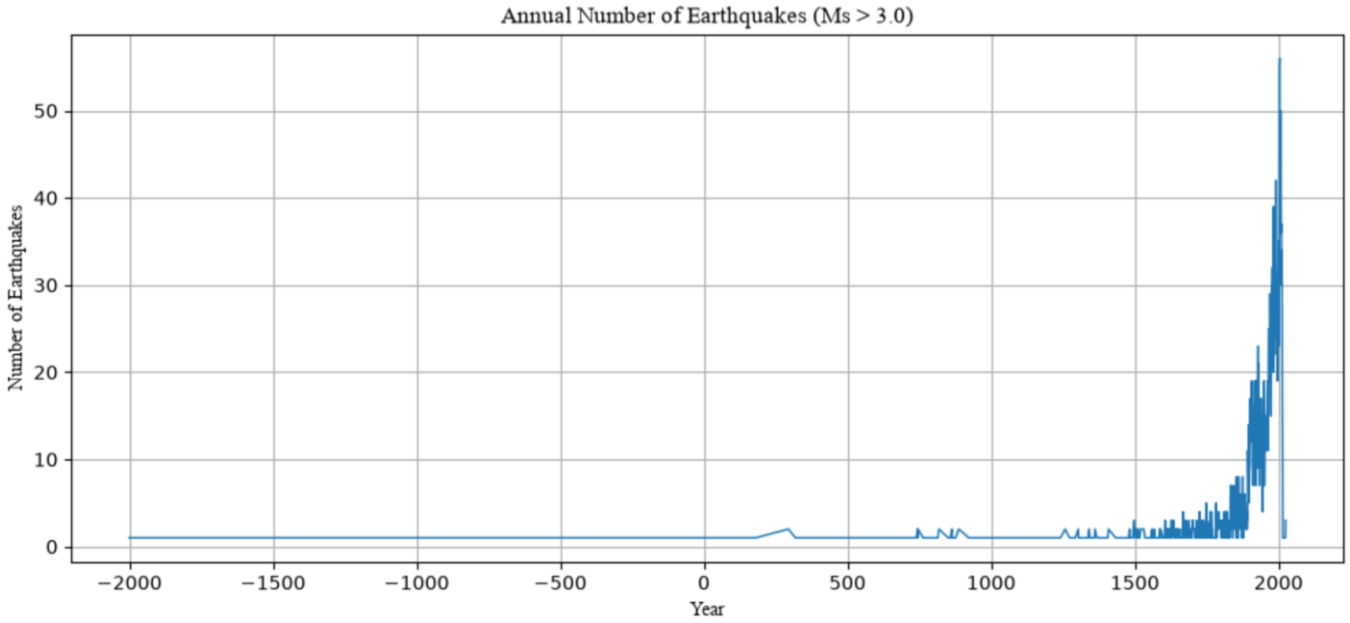


Figure 1 Annual Number of Earthquakes since 2500 B.C. Line Chart

1.3

The detailed results are provided in the output of the PS_1 1.3 code in the PS2.ipynb

2. Air temperature in Shenzhen 1998-2022

Based on the information provided in the user guide, data with quality codes of 1, 2, 4, and 5 were selected, and values greater than 618 or less than -932 were excluded. The remaining values were then divided by 10 to obtain the actual air temperature in degrees Celsius. The resulting monthly average air temperatures are presented in Figure 2.
Monthly air temperatures exhibit a cyclical, wave-like pattern, with higher temperatures in summer and lower temperatures in winter.

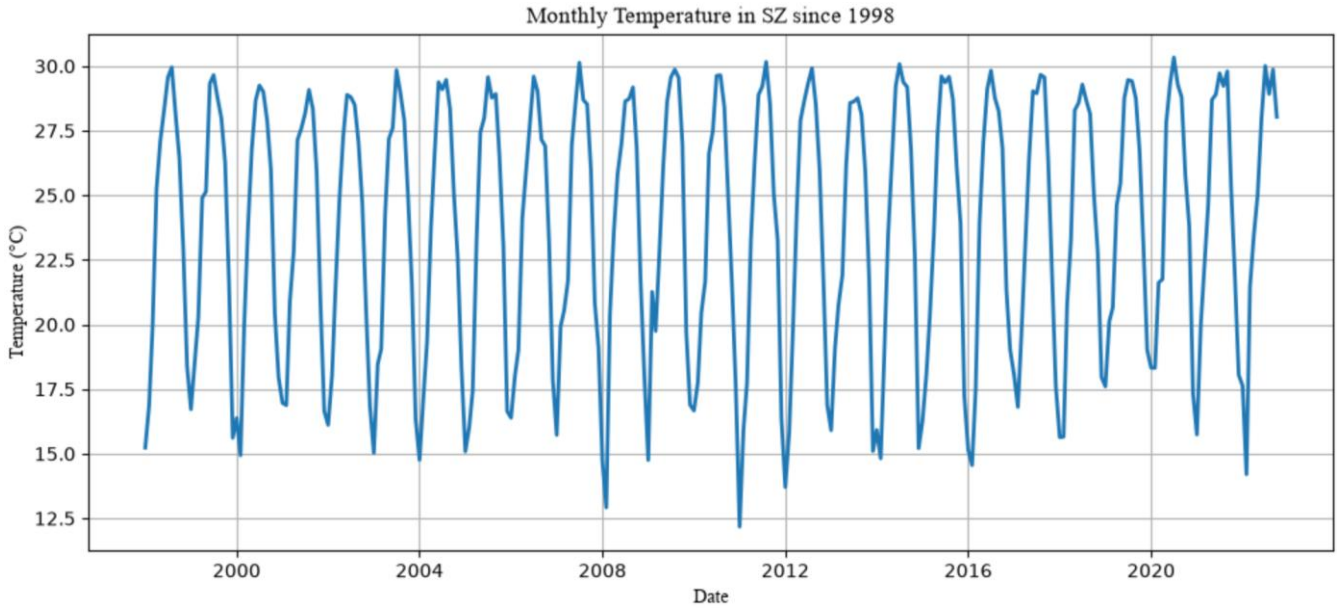


Figure 2 Monthly Averaged Air Temperature in SZ since 1998 Line Chart

3. Global collection of hurricanes

3.1

Table 2 The Top 10 largest hurricanes

	Name
1	ABLE
2	CHARLIE
3	EASY
4	FOX
5	GEORGE
6	CHARLIE
7	EASY
8	FOX
9	BAKER
10	CHARLIE

3.2

1. 95m/s
2. 95m/s
3. 95m/s
4. 95m/s
5. 95m/s
6. 95m/s
7. 95m/s
8. 95m/s
9. 95m/s
10. 95m/s
11. 95m/s
12. 95m/s
13. 95m/s
14. 95m/s
15. 95m/s
16. 95m/s
17. 95m/s
18. 95m/s
19. 95m/s
20. 95m/s

3.3

Table 3 Datapoints in Each Basin

Basin	Datapoints
EP	62412
NI	55401
SA	119
SI	160668
SP	67119
WP	236576

3.4

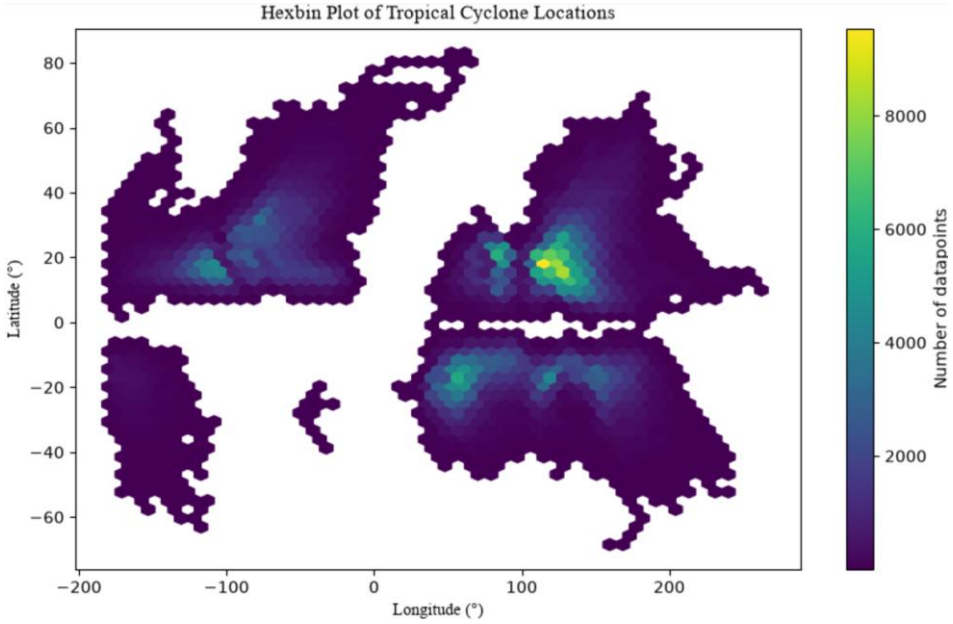


Figure 3 Hexbin Plot of the Location of Datapoints

3.5

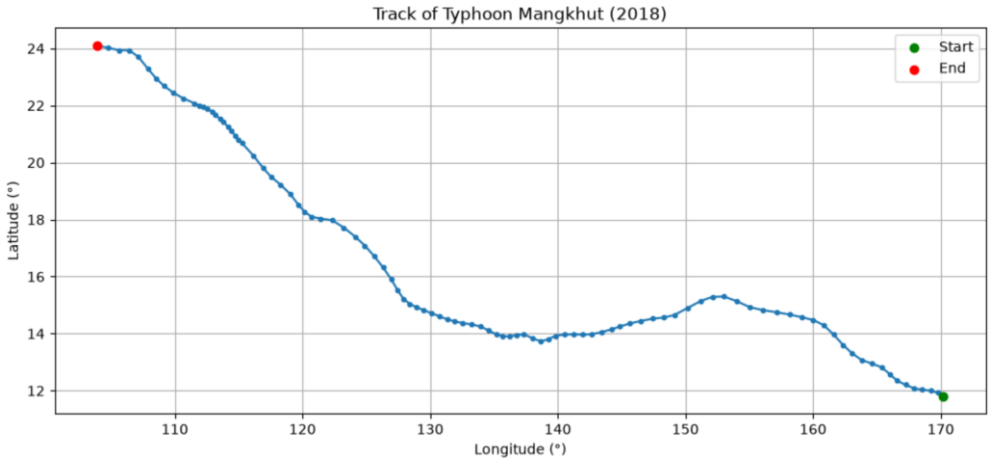


Figure 4 Track of Typhoon Manghut (2018)

3.6

The detailed results are provided in the output of the PS_3 3.6 code in the PS2.ipynb

3.7

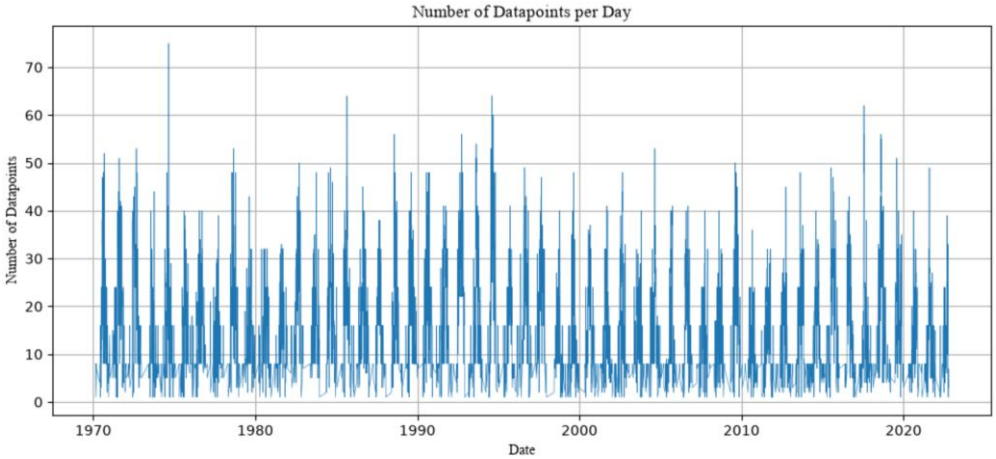


Figure 5 Number of Datapoints per Day Line Chart

3.8

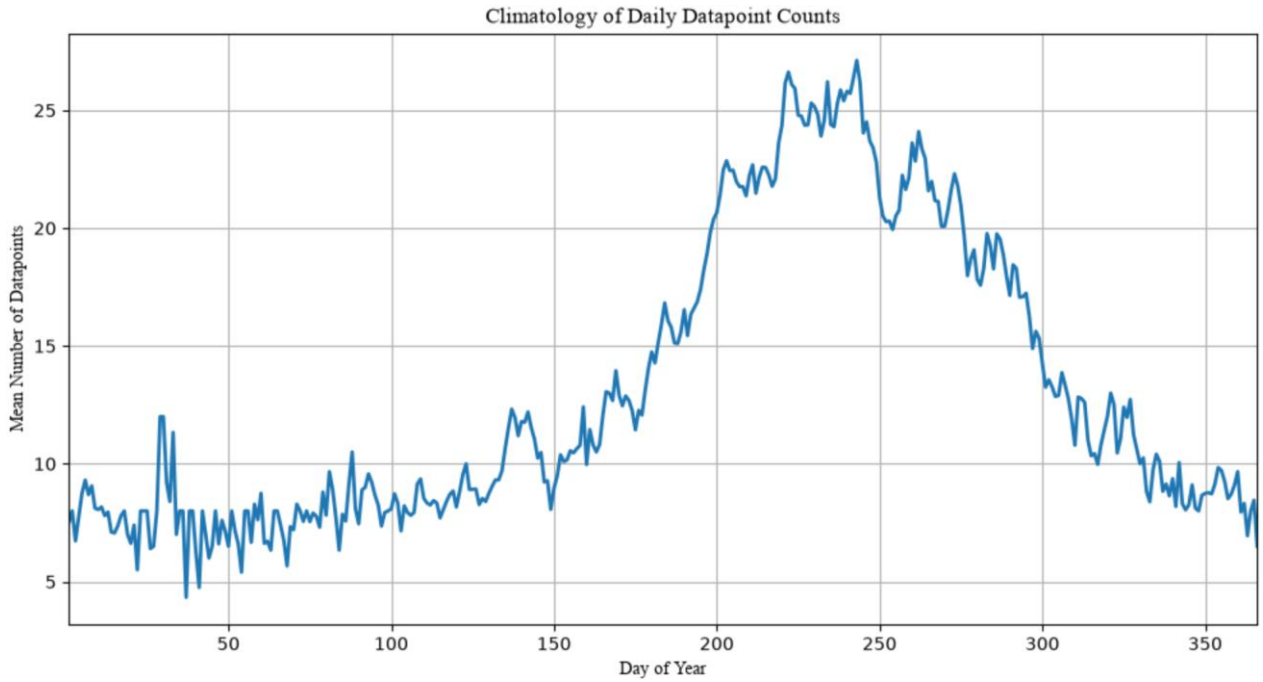


Figure 6 Climatology of Datapoints Counts Line Chart

3.9

The detailed results are provided in the output of the PS_3 3.9 code in the PS2.ipynb

3.10

Based on the Figure 7, several years stand out as having anomalously high or low hurricane activity, measured by the annual number of datapoints.

Unusually high activity: 1992 (approximately 5,000 datapoints) and 1994 (approximately 4,950) are the most prominent peaks. Other high-activity years include 1971 and 2015 (both around 4,450), 1997 (around 4,550), and 1990 (around 4,450).

Unusually low activity: 2010 is the clearest outlier, with only about 1,700 datapoints. Other notably low years include 1973, 1977, and 1998, each with approximately 2,300–2,450 datapoints, as well as 2022 (around 1,900).

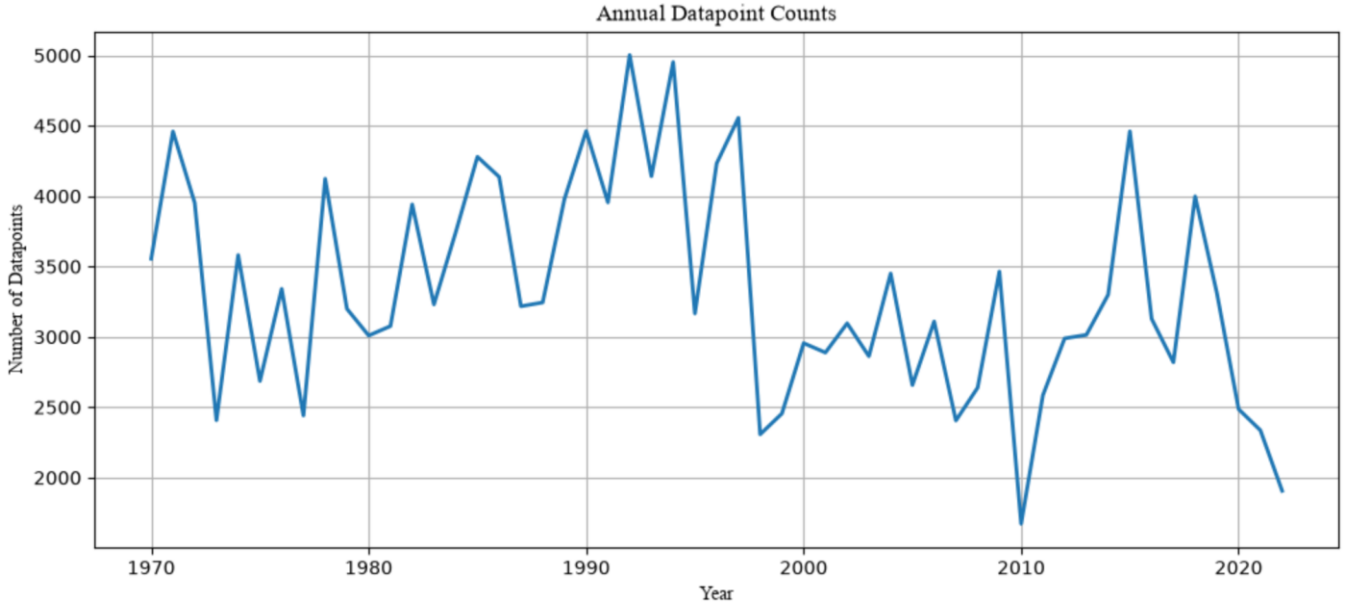


Figure 7 Annual Datapoints Counts

4. Explore a data set

4.1

The detailed results are provided in the output of the PS_4 4.1 code in the PS2.ipynb

4.2

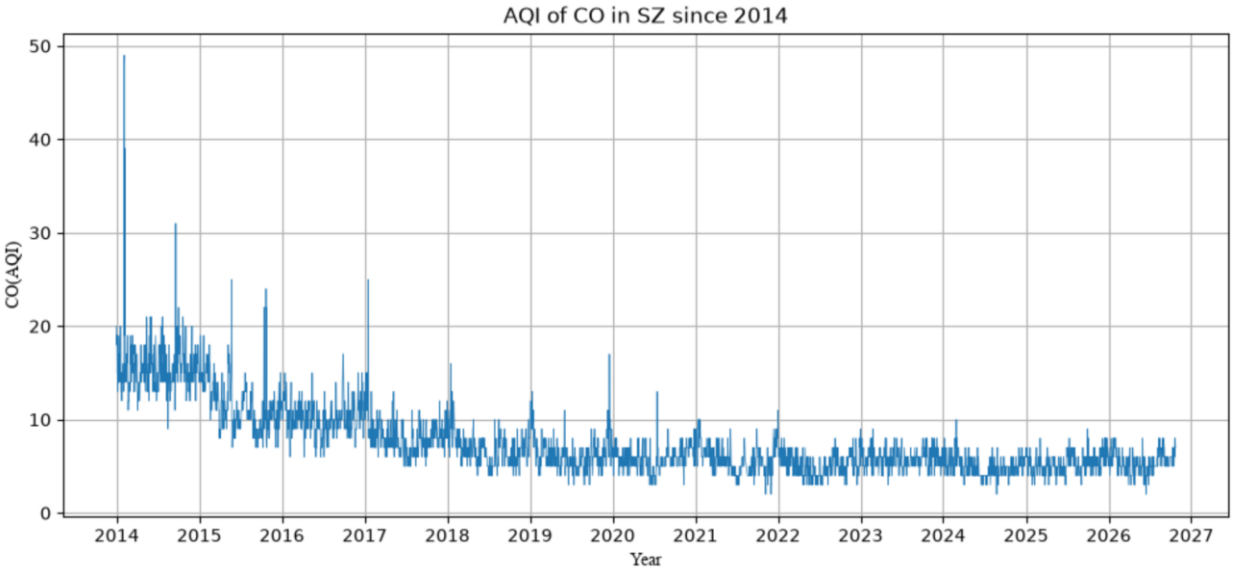


Figure 8 AQI of CO in SZ since 2014 Line Chart

4.3

Based on the CO-based AQI data, descriptive statistics, skewness, distribution characteristics, outliers, and interannual variations were analyzed.

4.3.1 Descriptive Statistics

Count: 4650.000000
Mean: 7.479785
Std: 3.618403
Min: 2.000000
25%: 5.000000
50%: 6.000000
75%: 9.000000
Max: 49.000000

4.3.2 Skewness

Median: 6.0
Skewness: 2.089452543038353

4.3.3 Distribution Characteristics

The AQI of CO distribution exhibits pronounced positive skewness, with most observations concentrated between 4 and 8 and the highest frequency (approximately 1,480) in the 4 - 5 interval. Most values range from 3 to 18, while a few exceed 25, including potential outliers near 50. The distribution is markedly asymmetric and may deviate from normality. Therefore, potential outliers and their effects on subsequent statistical analyses should be assessed.

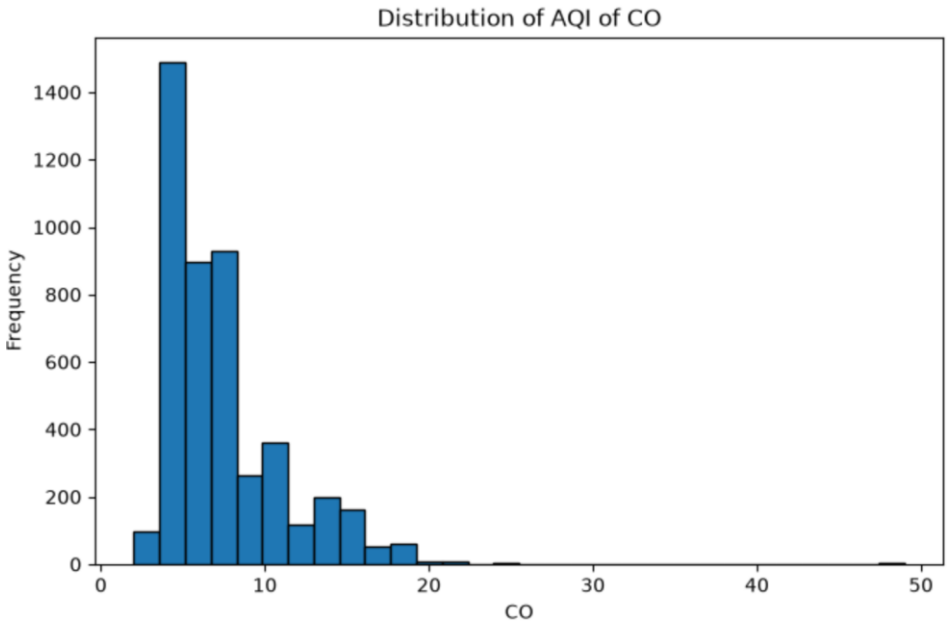


Figure 9 Distribution of AQI of CO

4.3.4 Outliers

Lower bound: -1.0

Upper bound: 15.0

Potential outliers: 209

Outlier percentage: 4.494623655913978%

4.3.5 Interannual Variations

The annual mean CO AQI exhibited a pronounced decline from 2013 to 2022, followed by fluctuations at relatively low levels through 2026. Specifically, it decreased from approximately 20 in 2013 to 8.2 in 2017, with the most substantial reduction occurring between 2013 and 2015. The decline then slowed, reaching approximately 5.2 in 2022. From 2022 onward, the annual mean fluctuated between approximately 5.0 and 5.8, with a slight increase in 2023, a further decline in 2024, and modest increases in 2025 and 2026.

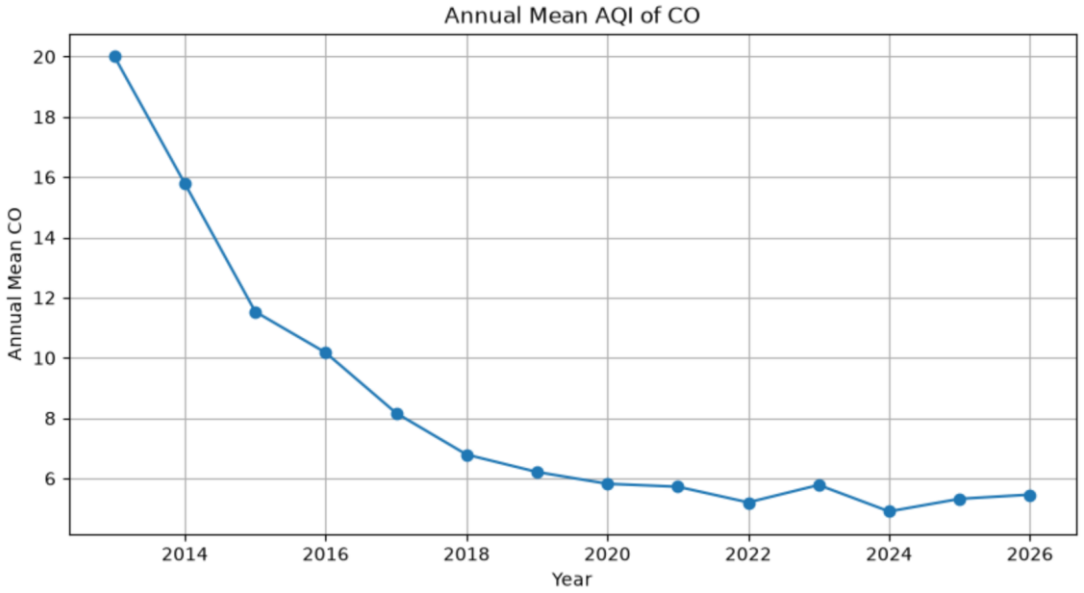


Figure 10 Annual Mean AQI of CO